Reactions of Acids and Bases

Neutralisation Reactions: Hydroxides and Oxides

A neutralisation reaction occurs when an acid reacts with a metal hydroxide or metal oxide to form water and a salt.

The water comes from the reaction between the H^+ in the acid and the OH^- produced by dissolving the hydroxide or the oxide in water. The salt is made from whatever is left.

For each of the word equations below, write the formula equation and then balance the equation.

1. Potassium hydroxide + Hydrochloric acid → Water + Potassium chloride

2. Nitric acid + Calcium hydroxide → Water + Calcium nitrate

3. Lithium oxide + Nitric acid → Water + Lithium nitrate

4. Sulphuric acid + Calcium oxide → Water + Calcium sulphate

5. Hydrochloric acid + Aluminium hydroxide → Water + Aluminium chloride

6. Nitric acid + Sodium hydroxide → Water + Sodium nitrate

7. Hydrochloric acid + Sodium oxide → Water + Sodium chloride

8. Nitric acid + Barium hydroxide → Water + Barium nitrate

9. Sulphuric acid + Magnesium oxide → Water + Magnesium sulphate

10. Calcium hydroxide + Hydrochloric acid→ Water + Calcium chloride

11. Aluminium oxide + Sulphuric acid → Water + Aluminium sulphate

12. Hydrochloric acid + Barium oxide → Water + Barium chloride

Neutralisation Reactions: Carbonates (CO_3^{-2}) and Hydrogencarbonates (HCO_3^{-1})

A neutralisation reaction occurs when an acid reacts with a carbonate or a hydrogencarbonate to form water, carbon dioxide and a salt.

The water comes from the reaction between the H^+ in the acid and an OH^- formed when the carbonate or hydrogen carbonate dissolves. The carbon dioxide comes from the carbonate. The salt is made from whatever is left.

For each of the word equations below, write the formula equation and then balance the equation.

1. Calcium carbonate + Hydrochloric acid → Water + Carbon dioxide + Calcium chloride

2. Nitric acid + Calcium carbonate → Water + Carbon dioxide + Calcium nitrate

3. Nitric acid + Sodium hydrogencarbonate → Water + Carbon dioxide + Sodium nitrate

4. Hydrochloric acid + Aluminium carbonate → Water + Carbon dioxide + Aluminium chloride

5. Calcium hydrogencarbonate + Hydrochloric acid → Water + Carbon dioxide + Calcium chloride

6. Nitric acid + Sodium carbonate → Water + Carbon dioxide + Sodium nitrate

7. Nitric acid + Barium carbonate → Water + Carbon dioxide + Barium nitrate

8. Nitric acid + sodium hydrogencarbonate → Water + Carbon dioxide + sodium nitrate

9. Sulphuric acid + Iron(III) carbonate → Water + Carbon dioxide + Iron(III) sulphate

10. Lead(II) carbonate + Hydrochloric acid→ Water + Carbon dioxide + Lead(II) chloride

11. Sulphuric acid + Barium hydrogencarbonate → Water + Carbon dioxide + Barium sulphate

12. Calcium hydrogencarbonate + Nitric acid → Water + Carbon dioxide + Calcium nitrate

Acid and Metals

When an acid and a metal are reacted together, hydrogen gas and a salt are produced.

The hydrogen gas comes from the H^+ in the acid. The salt is made from whatever is left.

For each of the word equations below, write the formula equation and then balance the equation.

1. Zinc + Hydrochloric acid → Hydrogen + Zinc chloride

2. Nitric acid + Calcium → Hydrogen + Calcium nitrate

3. Hydrochloric acid + Aluminium → Hydrogen + Aluminium chloride

4. Nitric acid + Iron → Hydrogen + Iron(II) nitrate

5. Nitric acid + Zinc → Hydrogen + Zinc nitrate

6. Sulphuric acid + Magnesium → Hydrogen + Magnesium sulphate

7. Iron + Hydrochloric acid→ Hydrogen + Iron(II) chloride